

Software Requirements Specification (SRS)

Digital Evidence Vault

1. Introduction

Digital Evidence Vault is a web application that allows users to securely upload, organize, search, download, and manage digital evidence such as images, videos, and PDF documents. The application is developed using React, Spring Boot, MySQL, JWT Authentication, and local file storage.

2. Objectives

- Provide secure user authentication.
- Allow users to upload digital evidence.
- Store files locally and metadata in MySQL.
- Enable viewing, searching, downloading, and deleting evidence.
- Provide a simple dashboard with evidence statistics.

3. Scope

The system supports user registration, login, evidence upload, evidence management, dashboard, profile management, and search functionality. The project is intended as a simple MVP.

4. Functional Requirements

FR1: User Registration
FR2: User Login using JWT Authentication
FR3: Upload Image, Video, or PDF
FR4: View Evidence List
FR5: Search by Title or Category
FR6: View Evidence Details
FR7: Download Evidence
FR8: Delete Evidence
FR9: View User Profile
FR10: Dashboard displaying total files and recent uploads

5. Non-Functional Requirements

- Secure password storage using BCrypt.
- JWT-based authentication.
- Responsive UI.
- Fast file upload/download.
- Simple and maintainable architecture.

6. System Architecture

Frontend: React
Backend: Spring Boot REST API
Database: MySQL
Storage: Local uploads/ folder
Authentication: JWT

7. Database Design

Users(id, name, email, password, role)

Evidence(id, title, description, filename, filepath, category, uploadDate, userId)

8. API Endpoints

POST /register

POST /login

GET /profile

POST /evidence/upload

GET /evidence

GET /evidence/{id}

DELETE /evidence/{id}

GET /download/{id}

9. Project Workflow

Register → Login → JWT Generated → Dashboard → Upload Evidence → Store File in uploads/ → Save Metadata in MySQL → View/Search/Download/Delete Evidence

10. Future Enhancements

Cloud storage integration, encryption, audit logs, OCR, AI-based search, secure sharing, and evidence verification.

11. Conclusion

The Digital Evidence Vault provides a simple, secure, and scalable platform for managing digital evidence using modern full-stack technologies.